Neerav Chaudhary

APPLIED AI ENGINEER | PRODUCT SYSTEMS, LLM EVALUATION, HEALTHCARE AI

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LinkedIn | GitHub | Portfolio

Applied AI engineer and published researcher building evaluated systems for industrial compliance, employee operations, healthcare, and scientific work. Takes products from workflow and interface design through full-stack implementation, controlled AI use, and measured operational outcomes.

SELECTED AI PRODUCTS

Relay - Voice-first employee operations | React, TypeScript, Fastify, Deepgram

- Built one authenticated voice-and-text workflow across People Ops, IT, Finance, and Administration, replacing the employee's need to locate the right portal, form, and service owner.
- Across 252 live conversations and 764 turns, completed 103/110 development workflows and 39/42 held-out rewordings; reached the correct record in 186/187 tool runs and made zero unsafe state changes across 21 red-team runs.

TraceLens - Healthcare AI safety and evaluation | Python, PyTorch, Hugging Face

- Built a preregistered evaluation and safeguard workspace spanning 277 saved cases, with physician-labelled data, hash-pinned models and prompts, frozen analysis, multi-GPU runners, and inspectable first-to-final action records.
- Vega caught 18/31 chatbot misses and disclosed every flag; reconsideration raised final emergency guidance from 39/70 to 47/70 while false alarms moved from 4/134 to 5/134, demonstrating a transparent support layer rather than a clinical-outcome claim.

Refract - Provenance-first research workspace | React, FastAPI, PostgreSQL, ChromaDB

- Unified Reader, Index, Compare, Review, and Stats for research and life-sciences teams, keeping page-level sources attached to RAG answers, synthesis, recall attempts, dataset audits, and model decisions.
- Passed 9/9 Reader and workspace checks, 8/8 recall-first scenarios, and 6/6 duplicate-repair cases; review gates stop analysis at leakage or data-quality issues before a result can be accepted.

Homeward - Source-grounded discharge coordination | React, Express, SQLite, MCP

- Built a transitions-of-care workflow that converts exact discharge instructions into recovery tasks, keeps missing or conflicting details visible, routes clarifications to a care-team view, and executes reminders only after patient approval.
- Passed 32/32 guardrail scenarios, 59 backend and contract tests, and four end-to-end UI journeys covering cross-patient access, prompt injection, approval bypass, duplicate actions, response loss, and provider outage on synthetic records.

PROFESSIONAL EXPERIENCE

NTPC | AI/ML Engineer — **Nov 2023-May 2024**

- Reduced regulatory and wrongful-citation risk by building an XGBoost + SHAP attribution model that distinguished plant-stack PM2.5 spikes from upwind traffic and regional background, producing legally defensible source evidence validated at r = 0.93-0.96 against ground truth (model R2 = 0.84).
- Protected generation continuity with an LSTM that forecast 24-hour exceedances at RMSE 9.9 ug/m3 for pre-emptive curtailment and a Random Forest virtual sensor that filled maintenance gaps at MAPE 4.8%, raising regulatory data uptime from 92.16% to 100%.
- Cut incident-to-report turnaround from approximately nine hours to 22 seconds by serializing SHAP outputs to JSON and generating internal briefs and regulator-ready letters with local Llama 3.1 on llama.cpp/Ollama plus a provider-agnostic fallback.
- Shortlisted AQMS station providers by mapping plant, ML-pipeline, and CPCB requirements to reference-grade CAAQMS (INR 1-2 crore per unit) and lower-cost IoT options, preventing over-spec cost and under-spec compliance gaps.

Deep Strat | Artificial Intelligence Research Intern — **Aug-Nov 2023**

- Partnered with department leadership to build a three-stage LLM workflow that converted raw customer complaints into sentiment- and intent-classified, correctly routed, SOP-grounded cases for operational review.
- Evaluated Random Forest and CNN models on 2,000 customer records; reduced false churn alarms 25% at stable AUC, decreasing unnecessary retention outreach without weakening discrimination.
- Advised founders and policy researchers on AI safety and guardrails; authored a 50-page technical reference that supported jointly authored research presented at the Nullcon security conference.

Manipal University Jaipur | AI Researcher, Medical Imaging — **Aug-Dec 2022**

- Co-authored an IEEE paper and built its Python/OpenCV atrophy pipeline across 2,198 ADNI MRI images; focusing measurement on the hippocampal region improved reported classification accuracy from 66% to 71%.
- Built a separate four-class TensorFlow experiment on 6,400 MRI images; probability-level ensembling improved held-out accuracy from 78% to 85%.

EDUCATION

San Jose State University | M.S., Industrial and Engineering Management | GPA 3.81 — **2024-2026**
Manipal University Jaipur | B.Tech., Computer Engineering | GPA 8.8/10 — **2018-2022**

PUBLICATIONS

N. Chaudhary and S. S. Verma, "Analyzing Brain Atrophy in Dementia Using Computer Vision," IEEE ICRITO, 2022.
doi:10.1109/ICRITO56286.2022.9964994

SKILLS

AI systems: LLM evaluation, voice agents, structured tool calling, RAG, multi-agent workflows, grounding, human-in-the-loop controls, Hugging Face, Deepgram, LangChain, llama.cpp, Ollama
Engineering: Python, TypeScript, React, FastAPI, Fastify, Express, SQL, PostgreSQL, SQLite, REST APIs, authentication and authorization, Docker, AWS, Git
ML and research: XGBoost, SHAP, LSTMs, Random Forest, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, forecasting, attribution, calibration, data-quality and leakage audits